

Coffee Shop Sales Analysis

Statistical Analysis Report

By: Ehab Talaat

1. Introduction and Objectives

This report presents a statistical analysis of transaction-level sales data from a coffee shop chain operating three locations in New York City (Astoria, Hell's Kitchen, and Lower Manhattan) between January and June 2023. The dataset comprises 149,116 individual transactions.

The analysis was structured around five research questions, each addressed with a hypothesis-driven statistical test appropriate to the data type and sample size involved.

- Q1 - Does revenue per transaction differ significantly across the three store locations?
- Q2 - Do weekday transactions generate significantly higher revenue than weekend transactions?
- Q3 - Does revenue per transaction differ significantly across product categories?
- Q4 - Does revenue per transaction differ significantly across time of day (morning, afternoon, evening)?
- Q5 - Is there a significant correlation between transaction quantity and revenue?

2. Data and Methodology

2.1 Data Source

Raw data was cleaned and transformed in Power Query. An engineered Revenue field ($\text{transaction_qty} \times \text{unit_price}$) was created, along with Day_of_Week, Is_Weekend, and Time_Period (Morning < 12:00, Afternoon 12:00–17:00, Evening > 17:00) fields.

2.2 Justification for Parametric Testing at Large Sample Size

With $n = 149,116$, formal normality tests such as Shapiro-Wilk are not appropriate, at this sample size, such tests reject the null hypothesis of normality for even trivially small deviations from a perfect normal shape, making the result uninformative. Instead, normality of the sampling distribution is justified by the Central Limit Theorem: every group compared in this analysis has n well in excess of 30 (the smallest group, Branded product category, has $n = 747$). Parametric tests (ANOVA, independent-samples t-test) are therefore statistically justified for all five research questions.

2.3 Test Selection Logic

Question	Test Used	Rationale
Q1 , Location (3 groups)	One-way ANOVA + pairwise t-tests	3+ groups; pairwise comparisons needed post-hoc
Q2 , Weekday vs Weekend (2 groups)	F-test for variance equality → Welch's t-test	2 groups; F-test determines equal/unequal variance form

Question	Test Used	Rationale
Q3 , Product Category (9 groups)	One-way ANOVA	3+ groups
Q4 , Time of Day (3 groups)	Pairwise t-tests, Bonferroni-corrected	3 groups (see methodological note, §4.4)
Q5 , Quantity vs Revenue	Pearson correlation	Two continuous variables

2.4 Effect Size and Correction Standards

- Effect size for ANOVA reported as η^2 ($SS_{\text{between}} / SS_{\text{total}}$): 0.01 small, 0.06 medium, 0.14 large (Cohen, 1988).
- Effect size for t-tests reported as Cohen's d where calculated: 0.2 small, 0.5 medium, 0.8 large.
- Where more than one pairwise comparison is drawn from the same family of tests, a Bonferroni-corrected significance threshold is applied: $\alpha_{\text{adjusted}} = 0.05 / k$, where k is the number of comparisons.

3. Descriptive Statistics

Statistic	Revenue	Quantity
Mean	\$4.69	1.44 units
Median	\$3.75	1 unit
Std. Deviation	\$4.23	0.54 units
Minimum	\$0.80	1 unit
Maximum	\$360.00	8 units
Skewness	41.75	0.80
Kurtosis	3359.75	0.82
n	149,116	149,116

⚠ Data quality concern: Revenue skewness (41.75) and kurtosis (3359.75) are extreme , far beyond what is typical even for right-skewed retail transaction data. A maximum single-transaction value of \$360 against a mean of \$4.69 indicates the presence of one or more extreme outliers or bulk/wholesale transactions. These were not investigated or removed in the current version of the analysis. Because all subsequent hypothesis tests operate on group means, and the affected extreme values are almost certainly concentrated in a small number of transactions, this is unlikely to overturn the directional conclusions below , but it should be investigated before this analysis is used to inform pricing or inventory decisions. Recommend filtering or separately reporting transactions above a reasonable ceiling (e.g. the 99.9th percentile) in the next iteration.

4. Results

4.1 Q1 , Revenue by Store Location

One-Way ANOVA

Source	SS	df	MS	F	p	F crit
Between Groups	1,289.09	2	644.55	36.09	< 0.001	2.996
Within Groups	2,663,152.38	149,113	17.86	,	,	,
Total	2,664,441.47	149,115	,	,	,	,

Verdict: Significant, $F(2, 149113) = 36.09, p < 0.001$

Effect size: $\eta^2 = 1,289.09 / 2,664,441.47 = 0.0005$, a negligible effect. Location explains less than 0.1% of variance in per-transaction revenue.

Group Means

Location	n	Mean Revenue/Txn	Total Revenue
Astoria	50,599	\$4.59	\$232,244
Hell's Kitchen	50,735	\$4.66	\$236,511
Lower Manhattan	47,782	\$4.81	\$230,057

Post-Hoc Pairwise Comparisons (Bonferroni $\alpha = 0.0167$)

Pair	t	df	p (two-tail)	Result
Astoria vs Hell's Kitchen	-2.484	74,047	0.013	Significant
Astoria vs Lower Manhattan	-11.401	95,052	< 0.001	Significant
Hell's Kitchen vs Lower Manhattan	-5.459	82,276	< 0.001	Significant

All three pairwise differences remain significant after Bonferroni correction. However, given the negligible overall effect size ($\eta^2 = 0.0005$), these differences , while statistically detectable at this sample size , are not practically meaningful. Mean revenue per transaction ranges only from \$4.59 to \$4.81 across locations. Note that Hell's Kitchen generates the highest total revenue (\$236,511) despite Lower Manhattan having the highest mean transaction value (\$4.81) , the two metrics answer different business questions and should not be conflated.

4.2 Q2 , Weekday vs Weekend Revenue

$H_0: \mu_{\text{weekday}} \leq \mu_{\text{weekend}}$ | $H_1: \mu_{\text{weekday}} > \mu_{\text{weekend}}$ (one-sided, pre-specified before testing)

F-Test for Equality of Variances

	Weekend	Weekday
Mean	\$4.69	\$4.68
Variance	18.79	17.51

	Weekend	Weekday
Observations	41,606	107,510
F	1.073	,
p (one-tail)	< 0.001	,

The F-test indicates significantly unequal variances ($p < 0.001$), so a Welch's t-test (unequal variances) is used.

Welch's t-Test (Two-Sample, Unequal Variances)

	Weekend	Weekday
Mean	\$4.69	\$4.68
df	73,327	,
t Stat	0.328	,
p (one-tail)	0.371	,
p (two-tail)	0.743	,

Verdict: Not significant, $t(73327) = 0.328$, $p = 0.371$

The data do not support the hypothesis that weekday transactions have higher average value than weekend transactions. Mean revenue per transaction is statistically indistinguishable between weekday (\$4.68) and weekend (\$4.69). Note this test concerns average value per transaction, not total revenue, total revenue is naturally higher on weekdays simply because there are more weekdays and more transactions in the sample.

4.3 Q3, Revenue by Product Category

One-Way ANOVA

Source	SS	df	MS	F	p	F crit
Between Groups	894,594.50	8	111,824.31	9,421.03	< 0.001	1.938
Within Groups	1,769,846.97	149,107	11.87	,	,	,
Total	2,664,441.47	149,115	,	,	,	,

Verdict: Significant, $F(8, 149107) = 9421.03$, $p < 0.001$

Effect size: $\eta^2 = 894,594.50 / 2,664,441.47 = 0.336$, a large effect. Product category alone explains 33.6% of variance in per-transaction revenue, making it by far the strongest predictor identified in this analysis.

Group Means

Category	n	Mean Revenue/Txn	Total Revenue
Coffee Beans	1,753	\$22.87	\$40,085
Branded	747	\$18.22	\$13,607
Loose Tea	1,210	\$9.27	\$11,214
Packaged Chocolate	487	\$9.05	\$4,408

Category	n	Mean Revenue/Txn	Total Revenue
Drinking Chocolate	11,468	\$6.31	\$72,416
Coffee	58,416	\$4.62	\$269,952
Tea	45,449	\$4.32	\$196,406
Bakery	22,796	\$3.61	\$82,316
Flavours	6,790	\$1.24	\$8,409

Group means suggest two distinct revenue tiers: high-value, low-volume categories (Coffee Beans, Branded, Loose Tea, Packaged Chocolate) and standard, high-volume categories (Coffee, Tea, Bakery, Drinking Chocolate, Flavours). Formal pairwise post-hoc testing across all 9 categories (36 possible pairs) was not conducted given the clear separation visible in group means and the overriding size of the omnibus effect; this is noted as a possible extension.

4.4 Q4 , Revenue by Time of Day

⚠ Methodological note: An omnibus one-way ANOVA across the three time periods was not run in the underlying workbook prior to the pairwise comparisons below. Best practice is to establish overall significance via ANOVA before conducting pairwise tests. The Bonferroni-corrected pairwise t-tests below are still valid as independent tests, but the report cannot cite a single omnibus F-statistic for this question the way it can for Q1 and Q3. This should be added in the next revision (Anova: Single Factor on Morning/Afternoon/Evening revenue columns).

Group Means

Time Period	n	Mean Revenue/Txn	Total Revenue	% of Total
Morning	81,751	\$4.75	\$388,289	55.6%
Afternoon	44,427	\$4.61	\$204,721	29.3%
Evening	22,938	\$4.61	\$105,803	15.1%

Pairwise t-Tests (Bonferroni $\alpha = 0.0167$)

Pair	t	df	p (two-tail)	Result
Morning vs Afternoon	6.360	126,176	< 0.001	Significant
Morning vs Evening	5.396	71,366	< 0.001	Significant
Afternoon vs Evening	-0.203	47,194	0.840	Not significant

Morning transactions have a significantly higher average value (\$4.75) than both afternoon (\$4.61, $p < 0.001$) and evening (\$4.61, $p < 0.001$) transactions. Afternoon and evening do not differ significantly from each other ($p = 0.840$). Morning also accounts for the majority of total revenue (55.6%), reflecting both higher transaction volume and higher average value during that period.

4.5 Q5 , Correlation Between Transaction Quantity and Revenue

⚠ Not completed in the current version. No correlation formula (CORREL, PEARSON, or the Data Analysis ToolPak Correlation tool) exists anywhere in the submitted workbook. Because Revenue is directly derived from Quantity (Revenue = Quantity $\times$ Unit Price), a meaningful positive correlation is expected by construction , but this cannot be reported as a finding until it is actually computed. To complete this: run

=CORREL(Cleaned_Data[transaction_qty], Cleaned_Data[Revenue]) for r, square it for r^2 , and use the t-distribution conversion ($t = r\sqrt{(n-2)/(1-r^2)}$), $p = T.DIST.2T(ABS(t), n-2)$ for the p-value. This is the only one of the five research questions without a supporting result in this report.

5. Limitations

- Extreme outliers in Revenue (skew = 41.75, max = \$360 vs. mean = \$4.69) were not investigated, filtered, or discussed. This should be resolved before the analysis informs any operational decision.
- Q5 (quantity–revenue correlation) was not completed.
- Q4 lacks a supporting omnibus ANOVA; conclusions rest on three independent pairwise comparisons only.
- With a sample size this large, even practically trivial differences reach statistical significance (see Q1, $\eta^2 = 0.0005$). Statistical significance is reported alongside effect size throughout this report specifically to guard against over-interpreting such findings.
- No formal post-hoc testing was performed across all 9 product-category pairs; conclusions for Q3 rest on visual inspection of group means rather than corrected pairwise tests.
- The dataset covers six months (Jan–Jun 2023) from three NYC locations only; findings should not be generalized beyond this scope, season, or market.

6. Conclusion

Of the four completed research questions, product category (Q3) is the dominant driver of per-transaction revenue by a wide margin ($\eta^2 = 0.336$), while store location (Q1), though statistically significant, has a negligible practical effect ($\eta^2 = 0.0005$). No significant difference was found between weekday and weekend average transaction value (Q2). Time of day shows a clear pattern: mornings significantly outperform both afternoon and evening in average transaction value, while afternoon and evening do not differ from each other (Q4). The quantity–revenue correlation (Q5) remains outstanding and should be completed before this report is considered final.